

Project Topic: Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

GROUP 19

Week 1 Progress Report

Group Members:

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Work done this week:

- Researched and understood our project topic/ problem statement
- Understood the kind of defects talked about in our problem statement (holes, stains, warps, knots, etc.)
- Understood the core workflow of the defect detection
- Researched about the classical texture descriptors used, particularly GLCM, LBP, and Gabor filters
- Started exploring suitable datasets and possible image-processing approaches for the project.
- Discussed a possible workflow that includes image preprocessing, texture feature extraction, defect detection, and defect localization.

Literature Review:

- The traditional techniques such as LBP and Gabor filter have been employed in fabric defect detection but lack adaptability.
- The use of deep learning and the YOLO network helped improve automatic defect detection and localization.

- There is a need for lightweight techniques to maintain a balance between accuracy and real-time performance.
- The proposed technique has enhanced YOLOv5s by CBAM, FEM and CloU, achieving higher accuracy with 102 FPS detection rate.

Work to be done next week:

- Complete the preparation of the data set and decide on the types of defects in the fabrics that will be studied.
- Implement image pre-processing techniques such as grayscaling, denoising, and contrast adjustment.
- Implement texture feature extraction using GLCM, LBP, and Gabor filtering and analyze their performance.
- Start building defect detection and localization system and analyze results.

Challenges Faced:

- Identification of various kinds of defect types in the fabric along with their occurrence in the images.
- Selection of appropriate dataset which provides sufficient diversity in terms of texture and type of defects in the fabrics.
- Understanding the difference between GLCM, LBP and Gabor filters along with finding a way of using these filters for defect detection.

References:

- Sage Journals:
<https://journals.sagepub.com/doi/full/10.1177/15589250241232153>